

Task 1: Machine Learning Fundamentals

1. What is Machine Learning?

A subset of AI where systems learn patterns from data instead of being explicitly programmed with fixed rules, and use those patterns to make predictions on new, unseen data.

2. AI vs ML vs Deep Learning

Artificial Intelligence (AI) is the broad field of creating systems that can perform tasks that normally require human intelligence. Examples include reasoning, decision-making,

speech recognition, and image understanding.

Machine Learning is a subset of AI. It allows computers to learn from data and improve their performance without being directly programmed for every task.

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers. Deep learning is particularly useful for complex tasks involving images, speech, natural language, and large datasets.

The relationship can be represented as:

Artificial Intelligence → Machine Learning → Deep Learning

For example, a voice assistant is an AI application. Machine learning can be used to understand the user's speech, while deep learning can be used to recognize complex speech and language patterns.

3. What is a dataset?

A structured collection of data (usually rows and columns) used to train and test a machine learning model — e.g., a table of students with their study hours, attendance, and pass/fail result.

4. Features and Labels

Features = the input variables used to make a prediction (e.g., Study Hours, Attendance).

Label (also called target) = the output/answer you're trying to predict (e.g., Pass/Fail).

5. Training Data vs Testing Data

Training data = the portion of the dataset used to teach the model patterns.

Testing data = a separate, unseen portion used to check how well the model actually learned — evaluates real performance, not memorization.

6. Supervised Learning

The model learns from labeled data — meaning each example already has the correct answer, and the model learns the relationship between features and that answer. E.g., predicting Pass/Fail using past student data.

7. Unsupervised Learning

The model works with **unlabeled** data — no correct answers given. It finds hidden patterns or groupings on its own. E.g., grouping customers into segments based on purchase behavior, without being told the groups beforehand.

8. Classification vs Regression

Classification = predicting a category/class (e.g., Pass or Fail, Spam or Not Spam).

Regression = predicting a continuous number (e.g., predicting exact marks, house price).

9. Real-world applications

Spam email detection, product recommendations (Netflix/Amazon), fraud detection in banking, medical diagnosis prediction, house price estimation.

CONCLUSION

Machine Learning enables computers to learn from data and improve predictions over time without being explicitly reprogrammed for every scenario. Understanding concepts like features, labels, and the difference between supervised/unsupervised learning and classification/regression

forms the foundation for building any ML model — skills directly applied in this week's mini project predicting student pass/fail outcomes.